

- Currently, large cap stock volume is inversely related to prices. The relationship could be due to the current investor sentiment (since the financial crises). Investors are very weary of the market and fear further sharp declines. A cash investment of a fixed dollar amount will purchase fewer shares in a rising market but more shares in a falling market. Redemption of a fixed dollar amount will require few shares be traded in a rising market and more shares be traded in a declining market. We expect that this trend will stay constant and may additionally become negative for small cap volumes until investor sentiment and overall market confidence increases.
- Our analysis did not uncover any relationships between volume levels and correlation. However, correlation still remains a favorite indicator for portfolio managers. We suggest readers experiment with alternative correlation measures such as log change and actual level. This may improve the accuracy of our volume forecast model.

FORECASTING COVARIANCE

In this section we discuss a technique to construct a short-term risk model based on our price volatility forecasting model and multi-factor model to estimate covariance. In Kissell and Glantz (2003) we provide a detailed process to construct a short-term trading risk model based on a principal component risk model and a GARCH volatility estimate. In this section we provide a more general process that can incorporate any risk model combined with any volatility estimate (e.g., the HMA-VIX approach).

This process is as follows:

Let,

$\bar{C}$ = covariance matrix constructed from our multi-factor model

D = diagonal matrix of historical volatilities (from risk model)

$\hat{D}$ = diagonal matrix of forecasted volatilities (e.g., HMA-VIX, Garch, etc.)

P = diagonal matrix of current prices

Step 1: Convert the covariance matrix C to a correlation matrix Rho by dividing by the corresponding volatility terms:

$$Rho = D^{-1}CD^{-1}$$